

RELATIONAL DATABASES

1. WHAT IS NORMALISATION?

Normalisation is a systematic process of organizing data in a relational database to reduce redundancy (duplicate data) and improve data integrity. It involves dividing large, complex tables into smaller, related tables and defining relationships between them using primary and foreign keys.

2. WHAT IS A TABLE IN 1NF?

- All column entries are atomic (single, indivisible values; no repeating groups or arrays).
- Each column contains values of a single data type.
- Each row is uniquely identifiable by a primary key.

3. WHEN IS A TABLE IN 2NF?

- It is already in 1NF.
- It has no partial dependencies (every non-key attribute must depend on the entire primary key, not just a part of a composite primary key).

4. WHEN IS A TABLE IN 3NF?

- It is already in 2NF.
- It has no transitive dependencies (no non-key attribute depends on another non-key attribute; non-key attributes must depend only on the primary key).

5. DEPENDENCY DIAGRAM & IDENTIFICATION (UNNORMALIZED/ 1NF)

Primary Key

Composite Primary Key: (INV_NUM, PROD_NUM)

Dependency Identification

Full Functional Dependency:

- $(INV_NUM, PROD_NUM) \rightarrow QUANT_SOLD$

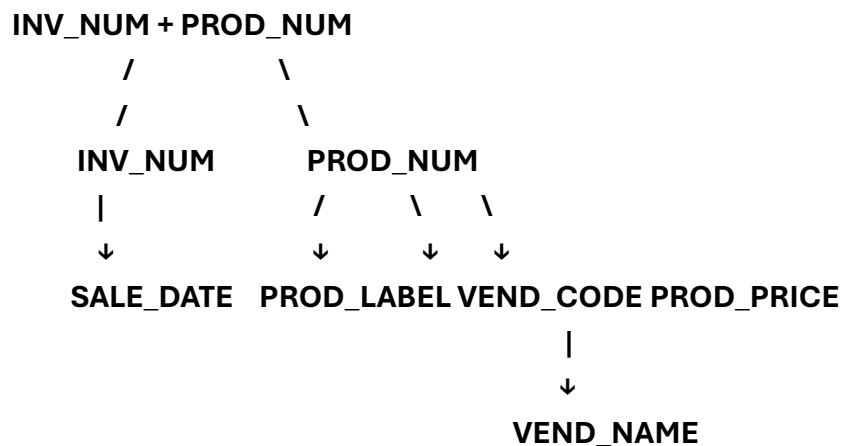
Partial Dependencies:

- $INV_NUM \rightarrow SALE_DATE$
- $PROD_NUM \rightarrow PROD_LABEL, VEND_CODE, VEND_NAME, PROD_PRICE$

Transitive Dependency:

- $VEND_CODE \rightarrow VEND_NAME$

DEPENDENCY DIAGRAM



6. REMOVING PARTIAL DEPENDENCIES (CONVERTING TO 2NF)

Table 1: INVOICE

- **Primary Key:** INV_NUM
- **Attributes:** INV_NUM, SALE_DATE
- **Dependency:** $INV_NUM \rightarrow SALE_DATE$

Table 2: PRODUCT

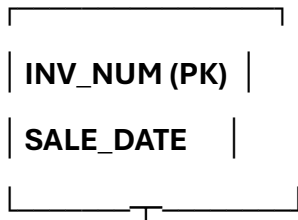
- **Primary Key:** PROD_NUM
- **Attributes:** PROD_NUM, PROD_LABEL, PROD_PRICE, VEND_CODE, VEND_NAME
- **Dependencies:**
 - $\rightarrow PROD_LABEL, PROD_PRICE, VEND_CODE, VEND_NAME$
 - $VEND_CODE \rightarrow VEND_NAME$

Table 3: INVOICE_LINE

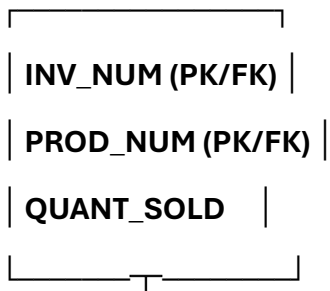
- **Composite Primary Key:** (INV_NUM, PROD_NUM)
- **Foreign Keys:** INV_NUM (references INVOICE), PROD_NUM (references PRODUCT)
- **Attributes:** INV_NUM, PROD_NUM, QUANT_SOLD
- **Dependency:** (INV_NUM, PROD_NUM) → QUANT_SOLD

DEPENDENCY DIAGRAM

INVOICE



INVOICE_LINE



PRODUCT



PROD_LABEL	
VEND_CODE	
PROD_PRICE	

7. REMOVING TRANSITIVE DEPENDENCIES (CONVERTING TO 3NF)

1. INVOICE table

- Primary Key: INV_NUM
- Attributes: INV_NUM, SALE_DATE
- Dependency: INV_NUM → SALE_DATE

2. VENDOR table

- Primary Key: VEND_CODE
- Attributes: VEND_CODE, VEND_NAME
- Dependency: VEND_CODE → VEND_NAME

3. PRODUCT table

- Primary Key: PROD_NUM
- Foreign Key: VEND_CODE (references VENDOR)
- Attributes: PROD_NUM, PROD_LABEL, PROD_PRICE, VEND_CODE
- Dependency: PROD_NUM → PROD_LABEL, PROD_PRICE, VEND_CODE

4. INVOICE_LINE table

- Composite Primary Key: (INV_NUM, PROD_NUM)
- Foreign Keys: INV_NUM (references INVOICE), PROD_NUM (references PRODUCT)
- Attributes: INV_NUM, PROD_NUM, QUANT_SOLD
- Dependency: (INV_NUM, PROD_NUM) → QUANT_SOLD

DEPENDENCY DIAGRAM

INV_NUM → SALE_DATE

PROD_NUM → PROD_LABEL

PROD_NUM → PROD_PRICE

PROD_NUM → VEND_CODE

VEND_CODE → VEND_NAME

INV_NUM + PROD_NUM → QUANT_SOLD